

PRAVIN KUMAR R

QUANTITATIVE DATA SCIENTIST

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PROFESSIONAL SUMMARY

Quantitative Data Scientist with an engineering analytical foundation (B.E. First Class Honors) and expertise in deep sequence forecasting, institutional climate risk modeling, and backtesting. Demonstrated success engineering multi-horizon neural architectures under walk-forward validation and automating Basel III & SEC Climate Value-at-Risk (VaR) waterfalls. Proficient across Python, SQL, PyTorch, financial statistics, and quantitative risk delivery.

SKILLS

Quantitative Risk: Multi-Horizon Forecasting, Basel III Climate VaR, Rolling Sharpe Ratio, Max Drawdown, Value-at-Risk (VaR)

ML & Deep Learning: PyTorch, PatchTST, Reversible Instance Normalization (RevIN), LSTMs, 1D-CNN, XGBoost, Scikit-Learn

Statistical Modeling & BI: Walk-Forward Validation (WFOV), Welch's t-Test, Mann-Whitney U, Power BI, Tableau, Stationarity, SHAP

Cloud, FinTech & Delivery: AWS, Azure, GCP, SQL, SQLite, SQLAlchemy, Docker, CI/CD (GitHub Actions), Streamlit, Git, Agile

PROJECTS

CryptoCast: Multi-Horizon Deep Sequence Forecasting

Tools: Python, PyTorch, PatchTST, LSTM, Streamlit, SHAP

- Accelerated quantitative risk hedging by architecting a multi-horizon sequence forecasting pipeline benchmarking 5 PyTorch models across **5,000+ records** under Walk-Forward Validation.
- Slashed 1-day price forecasting error to **2.06% MAPE** (3.85% for 3-day horizons) by deploying PatchTST and LSTM architectures with Reversible Instance Normalization (RevIN) to eliminate market drift.
- Drove a **\$92 MAE prediction error reduction** and guided hedging allocations by engineering multi-horizon lag features, on-chain dynamics, and an 11-page Streamlit analytics studio with SHAP explainability.

Alveris-Geo-AI: Institutional GeoAI & Basel III Climate VaR Engine

Tools: PyTorch, CNN, Copernicus DEM, InSAR, Streamlit

- Boosted pre-visual hazard detection to **95.98% zoning accuracy (+15.02% over RGB)** by developing a 13-band Sentinel-2 PyTorch CNN trained on Level-2A reflectance tensors.
- Replaced planar buffering with breach hydraulics by formulating an 8-way morphological flood routing algorithm on Copernicus 30m DEM and InSAR subsidence trajectories.
- Protected loan portfolios against impairment by automating a Basel III Climate Value-at-Risk (VaR) haircut waterfall translating physical hazards into dynamic collateral haircuts.

Other Verified Projects: Nifty 50 Stock Analysis (SQLAlchemy, Plotly), OrbitalComm Space Telemetry (95% SLA Failure Recall), SentixAI – on GitHub.

PROFESSIONAL EXPERIENCE

Quantitative Data Science Intern — Internship Studio

Jan 2026 – Feb 2026 | Remote

- Delivered **99.8% feature completeness** across **14,100+ session wearable records** by designing health analytics and data warehousing pipelines under 2-week Agile sprints.
- Reduced caloric prediction error by **89.2% over baseline models** (achieving an **R² of 0.9979** and **MAE of 3.78 kcal**) by developing a high-precision Random Forest regression model capturing nonlinear metabolic exertion.
- Cut executive reporting turnaround time by **75%** by deploying an interactive Streamlit and BI dashboard backed by K-Means and PCA clustering users into 4 exertion zones.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna University

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, ML for Earth Observation (EO College), Machine Learning & Pandas (HCL GUVI)

Additional: Stress Management (NPTEL)